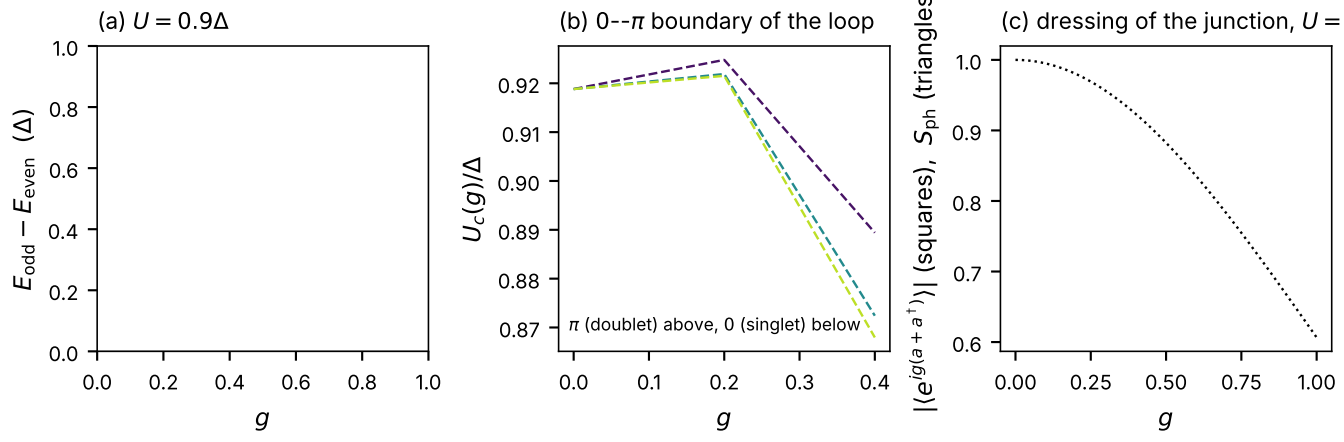
(a) $U = 0.9\Delta$
$E_{\mathrm{odd}} - E_{\mathrm{even}}$ (Δ)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
g
(b) 0--π boundary of the loop
$U_c(g)/\Delta$
0.92
0.91
0.90
0.89
0.88
0.87
π (doublet) above, 0 (singlet) below
0.0
0.1
0.2
0.3
0.4
g
(c) dressing of the junction, $U =$
$|\langle e^{ig(a+a^\dagger)}\rangle|$ (squares), S_{ph} (triangles
1.0
0.9
0.8
0.7
0.6
0.00
0.25
0.50
0.75
1.00
g